TOWARD THE TWIN PRIME CONJECTURE: CERTIFICATES, LIMITS OF THE SIEVE, AND A WORKING PROGRAMME

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Abstract. This is a living working paper. It does *not* prove the twin prime conjecture, and it does not improve the best known bound on small gaps between primes. It records (i) the state of the problem as of the date above, (ii) what this project has verified for itself, by exact computation and by the Lean 4 kernel, now including the established bound $H_1 \le 246$ of Polymath8b, reproduced end to end through a certified lower bound for $M_{50,1/25}$, (iii) small results of our own: a no-go statement for a natural attempt to sharpen the pigeonhole step, on the plain and on the ε-enlarged simplex; a generating-function formula that makes certified computation with all symmetric test functions cheap; an upper bound showing that the ε-enlargement of Polymath8b cannot, by itself, reach any $k \le 29$; numerical evidence that this upper bound is close to the limit of every dual weight depending only on t_m and $t_1 + \cdots + t_k$, and heuristic evidence that the ε-enlarged simplex stops at $k = 50$; and, for the one-band hybrid supports of Stadlmann, a rigorous floor (no $k \le 28$ at his level of distribution, whatever the caps), and an exact formula for his capped functionals, which gives a certified capped ratio $kJ_T/I_T = 4.0120 > 4$ at $k = 49$, computed in exact arithmetic by two independent programs: the variational step of Stadlmann's $H_1 \le 240$ is thereby reproduced, while his new equidistribution estimates, on which 240 rests, have not been verified by us, and (iv) a programme of original directions that later versions will test. Every statement carries one of the labels Lean-checked, Computed, Proved here, Literature, Claimed (a preprint not yet independently verified), or Open.

1. The problem and the honest position

Write p_n for the n-th prime and $H_1 = \liminf_{n\to\infty}(p_{n+1} - p_n)$. The twin prime conjecture is the statement $H_1 = 2$. It is Open.

A finite set $\mathcal{H} = \{h_1 < \cdots < h_k\}$ of integers is *admissible* if for every prime p it misses at least one residue class modulo p. Let DHL$[k, 2]$ be the statement that every admissible k-tuple has infinitely many translates $n + \mathcal{H}$ containing at least two primes, and let $H(k)$ be the least diameter $h_k - h_1$ of an admissible k-tuple. Then

$$\text{DHL}[k, 2] \implies H_1 \le H(k), \tag{1}$$

and DHL$[2, 2]$ applied to $\{0, 2\}$ is the twin prime conjecture. All progress on H_1 since 2013 consists of proving DHL$[k, 2]$ for smaller k.

Where the bound stands (12 November 2026).

Date: Version 0.6, 12 November 2026.

Prepared with the assistance of Claude (Anthropic). The computations, certificates and Lean sources accompany this paper.

$H_1 \leq$	k	Source	Label
7×10^7	3.5×10^6	Zhang [20]	Literature
600	105	Maynard [10]	Literature
246	50	Polymath8b [14]	Literature
240	49*	Stadlmann [18]	Claimed†
236	48*	Kintali [8]; Song–Yue [17]	Claimed
212	45	Charton et al. [3]	Claimed
186	40	OpenAI [13]	Claimed
182	39	GPT-6 Astra, prompted by Sharifi [15]	Claimed
12	5	Maynard, assuming EH [10]	Literature
6	3	Polymath8b, assuming GEH [14]	Literature

* Value of k inferred by us from $H(49) = 240$ and $H(48) = 236$, not read from the source. † Its variational step is reproduced here (Corollary 27); its equidistribution input is not verified by us.

The bound 246 stood from 2014 until the last days of August 2026. It has been formalised in Lean 4, assuming only the Bombieri–Vinogradov theorem [1]. The bounds from 240 to 186 appeared within one week. They rest on Stadlmann's idea of enlarging the support of the sieve weights using equidistribution estimates for moduli with a large smooth factor. The Lean development accompanying the bound 186 is conditional on three stated inputs (two exponential-sum bounds and a family of numerical integral bounds) [13]. An independent re-run of that repository, dated 7 and 8 November 2026, reproduced the build and the numerical certificate and describes the result as "conditionally supported, with the computation independently reproduced; not yet settled"; the two exponential-sum bounds remain axioms [12]. A further document, dated 18 September 2026 and naming an AI system as its author, claims DHL$[39, 2]$ and hence 182; by its own abstract its Lean development verifies the last implication "under explicit numerical and finite-field premises" [15]. We had missed it in earlier versions. (These details were read by an earlier run of this project; on 12 November the file was too large for our reader and direct download was refused, so we have not re-checked them since.) As of 7 September 2026 a standard reference still listed 246 as the best fully established value [11], and the page read the same on 12 November. On the date of this version we found no report of complete independent verification of any of the 2026 claims and no claimed bound below 182. We therefore keep three numbers apart throughout: *established* (246), *claimed* (182), and *reproduced by this project*: 246 unconditionally, and since version 0.6 also 240, conditionally on the equidistribution estimates of [18], which we have not verified (Corollary 27).

Why 2 is a different problem. Under the generalised Elliott–Halberstam conjecture the sieve gives $H_1 \leq 6$, and Polymath8b showed that 6 cannot be improved by purely sieve-theoretic arguments: this is Selberg's parity obstruction [14]. Lowering the bound (Track A below) and proving the conjecture (Track B) are therefore different tasks. No amount of success on Track A reaches 2.

2. What this project has verified for itself

2.1. **The combinatorial half, checked by the Lean kernel.** Our Lean development is written in core Lean 4, without Mathlib. It defines primality, the twin prime conjecture, the predicate `BoundedGaps B` (infinitely many pairs of distinct primes at distance at most B), admissibility, and DHL$[k, 2]$. It contains no `sorry`, and the only axioms it uses are `propext` and `Quot.sound`. The statement DHL$[k, 2]$ is always a hypothesis, never an axiom.

Theorem 1 (Lean-checked). (1) *The twin prime conjecture is equivalent to* `BoundedGaps 2`.

(2) *If a finite Boolean check succeeds on a list* $\mathcal{H}$ *(it misses a class modulo every* m *with* $2 \le m \le |\mathcal{H}|$*), then* $\mathcal{H}$ *is admissible. The moduli* $p > |\mathcal{H}|$ *are handled by a formal pigeonhole argument.*

(3) *For each pair* (k, D) *in Table 1, for* $(k, D) = (56, 278)$, $(57, 282)$, $(58, 288)$, $(59, 300)$, $(60, 304)$, $(61, 310)$, $(62, 320)$, $(63, 324)$, $(64, 330)$, *and for* $(102, 578)$ *and* $(105, 600)$, *there is an explicit admissible* k*-tuple in* $[0, D]$, *and* $\mathrm{DHL}[k, 2]$ *implies* `BoundedGaps D`. *In particular* $\mathrm{DHL}[2, 2]$ *implies the twin prime conjecture.*

(4) $\mathrm{DHL}[k, 2]$ *implies* $\mathrm{DHL}[k', 2]$ *for every* $k' \ge k$ *(`DHL.mono`: delete shifts from the tuple). So a proof of* $\mathrm{DHL}[j, 2]$ *with* $j \le k$ *gives* `BoundedGaps D` *for each listed pair* (k, D).

(5) `BoundedGaps 1` *is false (`not_boundedGaps_one`), and a pair* $\{0, d\}$ *is admissible if and only if* d *is even (`admissible_pair_iff`). So* 2 *is the smallest bound of this form that can hold, and the twin prime conjecture is the strongest statement of the form* `BoundedGaps D`.

Table 1. Pairs (k, D) with a kernel-checked admissible k-tuple of diameter D. The tuples were found by our own search; each D agrees with the value of $H(k)$ in published tables, whose optimality we have not re-proved.

k	D	k	D	k	D	k	D	k	D	k	D
2	2	11	36	20	80	29	130	38	176	47	226
3	6	12	42	21	84	30	136	39	182	48	236
4	8	13	48	22	90	31	140	40	186	49	240
5	12	14	50	23	94	32	146	41	188	50	246
6	16	15	56	24	100	33	152	42	196	51	252
7	20	16	60	25	110	34	156	43	200	52	254
8	26	17	66	26	114	35	158	44	210	53	264
9	30	18	70	27	120	36	162	45	212	54	270
10	32	19	76	28	126	37	168	46	216	55	272

Theorem 1 is elementary. Its value is that implication (1) is now a checked interface: any future proof of $\mathrm{DHL}[k, 2]$, by anyone, yields the corresponding gap bound with nothing left to trust on the combinatorial side.

2.2. **The sieve constant, with exact certificates.** Let $\mathcal{R}_k = \{t \in [0,1]^k : \ t_1 + \cdots + t_k \le 1\}$ and, for square-integrable F supported on $\mathcal{R}_k$,

$$I_k(F) = \int_{\mathcal{R}_k} F^2, \qquad J_k^{(m)}(F) = \int \left(\int F \, dt_m \right)^2 \prod_{i \ne m} dt_i, \qquad M_k = \sup_F \frac{\sum_{m=1}^k J_k^{(m)}(F)}{I_k(F)}.$$

Maynard's theorem [10, Prop. 4.2] states that if the primes have level of distribution θ and $\theta M_k/2 > 1$ then $\mathrm{DHL}[k, 2]$ holds. With Bombieri–Vinogradov ($\theta < 1/2$) the condition is $M_k > 4$.

We take $F = \sum_{a+2b \le d} c_{a,b} (1 - P_1)^a P_2^b$ with $P_j = \sum_i t_i^j$, as in [10, §8]. All integrals reduce to

$$\int_{\mathcal{R}_k} (1 - P_1)^A P_2^B \, dt = \frac{A!}{(k + A + 2B)!} \, G_k(B), \qquad \sum_{B \ge 0} G_k(B) \frac{x^B}{B!} = \left(\sum_{b \ge 0} \frac{(2b)!}{b!} x^b \right)^k,$$

which follows from the Dirichlet integral. The two quadratic forms are then exact rational matrices. We compute an approximate top eigenvector in floating point, round it to rationals, and evaluate the Rayleigh quotient in exact arithmetic; the last step is a rigorous lower bound whatever happened before it.

Theorem 2 (Computed, exact rational arithmetic)**.** *With F a polynomial in $1-P_1$ and P_2 only,*

$$M_{105} \geq 4.00206976, \qquad M_{102} \geq 4.00271343, \qquad M_5 \geq 2.00410541.$$

The first value reproduces Maynard's $M_{105} > 4$ with $d = 11$ (42 coefficients). The third gives DHL$[5,2]$ and $H_1 \leq 12$ under the Elliott–Halberstam conjecture, again reproducing [10]. For $d = 11, 15, 19$ the least k at which this family certifies $M_k > 4$ is $105, 102, 102$, and at $k = 101$ it gives only 3.99931 ($d = 19$): polynomials in P_1, P_2 alone stop at $k = 102$.

2.3. **All symmetric test functions.** To go further we allow every symmetric polynomial. Since $P_1, \dots, P_k$ generate the symmetric polynomials, the functions

$$(1-P_1)^a P^b := (1-P_1)^a \prod_{j \geq 2} P_j^{b_j}, \qquad a + w(b) \leq d, \quad w(b) = \sum_j j b_j,$$

span all symmetric polynomials of degree at most d (for $k \geq d$). The integrals have a closed form that we have not seen stated in this way (Proved here; possibly folklore).

Lemma 3. *Let $X_1, \dots, X_k$ be independent exponential random variables of mean 1 and $E_k(b) = \mathbb{E} \prod_j P_j(X)^{b_j}$. Then*

$$\int_{\mathcal{R}_k} (1-P_1)^A P^b \, dt = \frac{A!}{(k+A+w(b))!} \, E_k(b), \qquad \sum_b E_k(b) \frac{s^b}{b!} = \left(\sum_c w(c)! \, \frac{s^c}{c!} \right)^k,$$

where $s^b = \prod_j s_j^{b_j}$ and $b! = \prod_j b_j!$. In particular each $E_k(b)$ is a positive integer, computed exactly by multiplying exponential generating functions.

Proof. For a monomial, the Dirichlet integral gives $\int_{\mathcal{R}_k} (1-P_1)^A t^\alpha \, dt = A!\,\alpha!/(k+A+|\alpha|)!$, and $\alpha! = \mathbb{E} X^\alpha$. Summing over the monomials of P^b, all of degree $w(b)$, gives the first identity. For the second, $\mathbb{E} \exp(\sum_j s_j P_j(X)) = \left(\mathbb{E} \exp(\sum_j s_j X_1^j) \right)^k$ by independence, and $\mathbb{E} \exp(\sum_j s_j X_1^j) = \sum_c \mathbb{E} X_1^{w(c)} \, s^c/c! = \sum_c w(c)! \, s^c/c!$, as formal power series. □

The J-functional reduces to the same lemma in $k-1$ variables: with $P_j = P_j' + t_k^j$,

$$\int_0^{1-P_1'} (1-P_1)^a P^b \, dt_k = \sum_{e \leq b} \binom{b}{e} \frac{a! \, w(e)!}{(a+w(e)+1)!} \, (1-P_1')^{a+w(e)+1} P'^{\,b-e}.$$

So both Gram matrices are exact rational matrices. We find an approximate top eigenvector by inverse iteration with a fixed shift above $\frac{k}{k-1} \log k$, which can only converge to the top eigenvalue. We take its floating-point midpoints as exact dyadic rationals and evaluate the Rayleigh quotient exactly. The working precision of the eigenvector matters only for speed. A separate script, `code/verify_cert.py`, rebuilds both Gram matrices from scratch and re-evaluates each stored coefficient vector in exact arithmetic.

Theorem 4 (Computed, exact rational arithmetic)**.**

$$M_{105} \geq 4.17427, \qquad M_{64} \geq 4.00363, \qquad M_{58} \geq 4.00927,$$
$$M_{56} \geq 4.00519, \qquad M_{55} \geq 4.00516, \qquad M_{54} \geq 4.00238.$$

The first two use all symmetric polynomials of degree 11 *and* 12 *(*195 *and* 272 *coefficients). The others use the subspace spanned by* $(1-P_1)^a P^b$ *with* b *supported on even* j*, as in* [14, §7]*, of degree* 16*,* 18*,* 20 *and* 23 *(*307*,* 471*,* 707 *and* 1236 *coefficients).*

Basis	degree	coefficients	least k with certified $M_k > 4$
$1-P_1$, P_2	19	110	102
all symmetric	12	272	64
even power sums	16	307	58
even power sums	18	471	56
even power sums	20	707	55†
even power sums	23	1236	54†

† Only this value of k was run at this degree; smaller k were not tried.

Corollary 5 (end-to-end within this project)**.** *Assuming the Bombieri–Vinogradov theorem and* [10, Prop. 4.2]*, both* Literature*, Theorems 4 and 1 give* $H_1 \le 270$*.*

This reproduces a known result and is not new. Polymath8b obtained $M_{54} > 4.00238$ with even-part polynomials of degree 23 [14, Thm. 3.9], and hence $H_1 \le 270$. Our certified value in the same space, $4.002386\ldots$, agrees with theirs to every digit they print, which is an external check on our pipeline; two independent runs of it (one from an earlier session) produced the same value. Corollary 5 is the best that the plain simplex gives with Bombieri–Vinogradov (see the next subsection); Corollary 7 below takes the next rung of the ladder $600 \to 270 \to 246 \to 240 \to 212 \to 186 \to 182$.

2.4. **What the plain method cannot do.** Polymath8b proved $M_k \le \frac{k}{k-1}\log k$ [14]. Hence (Computed from that bound) $M_k > 4$ forces $k \ge 51$, and on the plain simplex DHL$[k,2]$ requires a level of distribution $\theta > 2(k-1)/(k\log k)$:

k	50	45	40	35	30	20	10	5
$\theta >$	0.501	0.514	0.529	0.547	0.568	0.634	0.782	0.994

The 2026 results go below $k = 50$ by changing the support of F, not by proving a uniform level of distribution of this size.

2.5. **Beyond the plain simplex: $H_1 \le 246$.** Polymath8b's ε-enlarged simplex [14, Thm. 3.12] (Literature; we re-read the statement in the arXiv text for this version) takes F supported on $(1+\varepsilon)\mathcal{R}_k$ and integrates the square of each marginal $\int F\,dt_m$ only over $(1-\varepsilon)\mathcal{R}_{k-1}$. Call the resulting constant $M_{k,\varepsilon}$. If EH$[\theta]$ holds with $1+\varepsilon < 1/\theta$ and $M_{k,\varepsilon} > 2/\theta$, then DHL$[k,2]$ holds. Bombieri–Vinogradov gives every $\theta < 1/2$, so for every fixed $\varepsilon < 1$ the condition is $M_{k,\varepsilon} > 4$.

We extended our pipeline to this functional, following the route that [14, §7.2] describes in words. Put $L = 1+\varepsilon$, $\tau = 1-\varepsilon$ and use the basis $(L-P_1)^a P^b$ (b supported on even j). The following identities are elementary (Proved here): the substitution $t = Ls$ and Lemma 3 give

$$\int_{L\mathcal{R}_k} (L-P_1)^A P^b\,dt = L^{k+A+w(b)}\,\frac{A!}{(k+A+w(b))!}\,E_k(b);$$

the inner integral over $t_m \in [0, L-P_1']$ is the formula displayed after Lemma 3 with $1-P_1'$ replaced by $L-P_1'$; and on $\tau\mathcal{R}_{k-1}$ we write $L-P_1' = 2\varepsilon + (\tau - P_1')$, so that

$$\int_{\tau\mathcal{R}_{k-1}} (L-P_1')^A P'^B = \sum_{i=0}^{A} \binom{A}{i} (2\varepsilon)^{A-i}\,\tau^{k-1+i+w(B)}\,\frac{i!}{(k-1+i+w(B))!}\,E_{k-1}(B).$$

For rational ε both Gram matrices are again exact rational matrices. We checked the code against hand computation for $k = 2, 3$ and $F \in \{1, L - P_1\}$, and against direct iterated symbolic integration (sympy, without Lemma 3 or the binomial rewrite) of every Gram entry for $(k, d, \varepsilon) = (3, 3, 1/5)$ and $(4, 4, 1/25)$: no mismatch (`code/check_eps_sympy.py`). At $\varepsilon = 0$ it is the old code path (an old certificate re-verifies unchanged). We also found that the working precision of the eigenvector search can be cut from about 6000 to 300–400 bits without changing any certified digit, which makes degree 25 feasible on a four-core machine (build 16 minutes, search 4 minutes).

Theorem 6 (Computed, exact rational arithmetic)**.** *With even power sums of degree* 25 *(*1780 *coefficients) and of degree* 27 *(*2526 *coefficients),*

$$M_{50,1/25} \geq 4.0012466 \qquad \textit{and} \qquad M_{50,1/25} \geq 4.0043063.$$

The certificates are `code/cert_M50e25_D25e.json` and `code/cert_M50e25_D27e.json` (the degree-27 run took 43 minutes to build and 10 minutes to search, with about 10 GB of memory); the script `code/verify_cert.py` rebuilds both Gram matrices from scratch and re-evaluates the stored rational vector exactly; it uses the same Gram-matrix builder, so it checks the arithmetic and the bookkeeping, while the formulas themselves are checked by the symbolic test above. Polymath8b report $M_{50,1/25} > 4.00124$ at degree 25 and > 4.0043 at degree 27 [14, §7.2]; our values agree with both to every printed digit.

Corollary 7 (end-to-end within this project)**.** *Assuming the Bombieri–Vinogradov theorem and* [14, Thm. 3.12(i)]*, both* Literature*, Theorem 6 and the Lean tuple* $(50, 246)$ *of Theorem 1 give* $H_1 \leq 246$.

This is the established bound of 2014, reproduced; it is not new. It closes the second rung of our ladder. Table 2 shows how the certified values grow with the degree and depend on ε; it is the data behind Remark 20 below.

Table 2. Certified lower bounds for $M_{k,\varepsilon}$ (Computed, exact Rayleigh quotients, even power sums of degree D). Data: `runs/scan_k49.jsonl`, `runs/scan_k50.jsonl`, written by `code/scan_eps.py`; the column $k = 50$, $D = 16$ from direct runs of `code/mk_general.py`. A dash: not computed.

	$k = 49$			$k = 50$		
ε	$D = 16$	$D = 18$	$D = 20$	$D = 16$	$D = 18$	$D = 20$
0	3.88640	3.90102	3.91056	3.90173	3.91718	3.92739
1/50	3.93491	3.95066	3.96110	3.94979	3.96644	3.97761
1/30	3.93969	3.95745	3.96933	–	3.97254	3.98522
1/25	3.93674	3.95563	3.96840	3.95041	3.97034	3.98395
1/20	3.92741	3.94823	3.96244	–	3.96234	3.97745

3. Results of our own: rigidity of the pigeonhole step

Every bound above ends with the same step: if the weighted average number of primes among $n + h_1, \ldots, n + h_k$ exceeds 1, two of them are prime, and they are at most $h_k - h_1$ apart. A natural hope is to do better with unequal weights. Fix D, and give each position a weight $c_i \geq 0$. Call a set of positions D-*separated* if any two of its shifts differ by more than D, and let $\alpha_D(c)$

be the largest total weight of a D-separated set. If no two primes among the $n+h_i$ are within D, then $\sum_i c_i \mathbf{1}_{n+h_i \text{ prime}} \le \alpha_D(c)$. Writing

$$M_k[c] = \sup_F \frac{\sum_m c_m J_k^{(m)}(F)}{I_k(F)},$$

the sieve therefore yields $H_1 \le D$ as soon as $\frac{\theta}{2} M_k[c] > \alpha_D(c)$. For $c = (1, \dots, 1)$ and $D = h_k - h_1$ this is the usual criterion. Could a cleverer c certify some $D < h_k - h_1$ that no sub-tuple certifies?

Proposition 8 (Proved here; possibly folklore)**.** *Let ω be the largest number of shifts h_i contained in an interval of length D. Then for every $c \ge 0$, not identically zero,*

$$\frac{M_k[c]}{\alpha_D(c)} \le M_\omega.$$

So the weighted criterion never certifies more than the standard criterion applied to the best sub-tuple of diameter at most D.

Proof. Let G be the graph on $\{1, \dots, k\}$ joining $i \ne j$ when $|h_i - h_j| \le D$. Its stable sets are the D-separated sets and its cliques are the sets of shifts lying in an interval of length D. Normalise $\alpha_D(c) = 1$, so c lies in

$$P = \{c \ge 0 : \ c(S) \le 1 \text{ for every stable set } S\}.$$

The graph G is a unit interval graph, hence perfect, and so is its complement by the perfect graph theorem [9]. For a perfect graph the polytope cut out by the clique inequalities is the convex hull of the indicator vectors of its stable sets [5]. Applying this to the complement of G gives

$$P = \operatorname{conv}\{\mathbf{1}_K : \ K \text{ a clique of } G\}.$$

The map $c \mapsto M_k[c]$ is a supremum of linear functionals, hence convex, and therefore $M_k[c] \le \max_K M_k[\mathbf{1}_K]$. It remains to show $M_k[\mathbf{1}_K] \le M_j$ for $|K| = j$; this suffices because M_j is non-decreasing in j (adjoin a variable and concentrate F near its zero set). Write $t = (s, u)$ with s the coordinates in K and u the others, put $\sigma(u) = 1 - \sum u \ge 0$, and let $F_u = F(\cdot, u)$, a function on the scaled simplex $\sigma(u)\mathcal{R}_j$. For $m \in K$ we have $J_k^{(m)}(F) = \int J_j^{(m)}(F_u)\, du$, and $I_k(F) = \int I_j(F_u)\, du$. Rescaling F_u to $\mathcal{R}_j$ multiplies I_j by σ^j and each $J_j^{(m)}$ by σ^{j+1}, so

$$\sum_{m \in K} J_j^{(m)}(F_u) \ \le \ \sigma(u)\, M_j\, I_j(F_u) \ \le \ M_j\, I_j(F_u).$$

Integrating over u gives the claim. □

Remark 9 (Computed, consistency check for $k = 3$). With unrestricted polynomials of degree 6 on $\mathcal{R}_3$ we find the lower bounds $M_3[(1,1,1)] \ge 1.6464$, $M_3[(1,1,0)] \ge 1.3579$ and $M_3[(\frac{1}{2}, 1, \frac{1}{2})] \ge 1.1604$, against $M_2 = 1/(1 - W(1/e)) = 1.3859\ldots$. The half-weighted functional is far below M_2, as the proposition requires.

The same rigidity on the enlarged simplex. The ε-enlarged simplex of Polymath8b [14, Thm. 3.12] takes F supported on $(1+\varepsilon)\mathcal{R}_k$ and restricts the outer integral of each $J^{(m)}$ to $(1-\varepsilon)\mathcal{R}_{k-1}$. Call the resulting constant $M_{k,\varepsilon}$. The criterion $M_{k,\varepsilon} > 2/\theta$ is valid in two cases (Literature): under EH[θ] when $1 + \varepsilon < 1/\theta$, and under GEH[$\theta$] when $\varepsilon < 1/(k-1)$. With Bombieri–Vinogradov the first case allows every $\varepsilon < 1$, and Polymath8b's $H_1 \le 246$ uses $M_{50,1/25} > 4.0043$ [14, Thm. 3.13]. Unequal weights are no more useful there.

Proposition 10 (Proved here; possibly folklore)**.** *For $0 \le \varepsilon < 1$ and every weight $c \ge 0$, $c \ne 0$, $M_{k,\varepsilon}[c]/\alpha_D(c) \le M_{\omega,\varepsilon}$, with equality for the indicator of a largest clique. Since the conditions on ε above do not get harder as k decreases, the weighted criterion never certifies more than the standard one for the best sub-tuple of diameter at most D.*

Proof. As for Proposition 8, it suffices to show $\sum_{m\in K} J^{(m)}_{k,\varepsilon}(F) \le M_{j,\varepsilon} I_k(F)$ for a clique K of size j, together with $M_{j,\varepsilon} \le M_{\omega,\varepsilon}$. Fix the coordinates u outside K, with $|u| \le 1-\varepsilon$ (other slices do not meet the J-region). The slice F_u is supported on $(1+\varepsilon-|u|)\mathcal{R}_j$, and its J-region is $(1-\varepsilon-|u|)\mathcal{R}_{j-1}$. Put $\mu = (1+\varepsilon-|u|)/(1+\varepsilon) \le 1$ and $H(x) = F_u(\mu x)$, which is supported on $(1+\varepsilon)\mathcal{R}_j$. Then $I_j(F_u) = \mu^j I_j(H)$, and each slice J-term equals μ^{j+1} times the integral of $(\int H\, dx_m)^2$ over $|x'| \le (1-\varepsilon-|u|)/\mu$. Since

$$\frac{(1-\varepsilon-|u|)(1+\varepsilon)}{1+\varepsilon-|u|} \le 1-\varepsilon \iff -|u|(1+\varepsilon) \le -|u|(1-\varepsilon),$$

that region lies inside $(1-\varepsilon)\mathcal{R}_{j-1}$, so the slice contributes at most $\mu^{j+1} M_{j,\varepsilon} I_j(H) = \mu M_{j,\varepsilon} I_j(F_u)$. For $j = 1$ use Cauchy–Schwarz, which gives $M_{1,\varepsilon} = 1+\varepsilon$. Integrate over u and use $\mu \le 1$. Monotonicity in j follows by embedding: take $F(s,u) = H(s/\mu_\delta)\mathbf{1}_{[0,\delta]^{k-j}}(u)$ with $\mu_\delta = 1-(k-j)\delta/(1+\varepsilon)$ and let $\delta \to 0$. □

Slicing naively gives each slice its own parameter $\varepsilon' = \varepsilon/(1-|u|) \ge \varepsilon$, which would only bound the weighted constant by $\sup_{\varepsilon' \ge \varepsilon} M_{j,\varepsilon'}$, a weaker statement. The single rescaling by μ above avoids that loss.

Remark 11 (Computed, $k = 3$). With unrestricted polynomials (degree 5 for $k = 3$, degree 10 for $k = 2$, cut off at $|t| \le \rho$ for $\rho \in \{1-\varepsilon, 1, 1+\varepsilon\}$):

ε	$M_{2,\varepsilon} \ge$	$M_{3,\varepsilon}[(1,1,0)] \ge$	$M_{3,\varepsilon}[(\frac{1}{2},1,\frac{1}{2})] \ge$	$M_{3,\varepsilon} \ge$
0	1.385933	1.349844	1.160121	1.646427
1/10	1.518567	1.482143	1.264698	1.793492
1/5	1.627161	1.585674	1.341926	1.882282
1/3	1.717104	1.663743	1.413309	1.917001

Both weighted columns stay below $M_{2,\varepsilon}$, as the proposition requires.

Remark 12 (scope). Neither proposition covers more general polytope supports [14, §3], Stadlmann's hybrid supports [18], J-regions that depend on m, or arguments that are not linear in the indicators of primality. Those are Open.

4. Results of our own: how far the ε-enlargement alone can go

For the plain simplex the bound $M_k \le \frac{k}{k-1}\log k$ shows where the method stops ($k \ge 51$). This section starts the same analysis for the ε-enlarged simplex. For $0 < \tau \le L$ put

$$M_k(L,\tau) = \sup_F \frac{\sum_{m=1}^k \int_{\tau\mathcal{R}_{k-1}} \left(\int_0^\infty F\, dt_m\right)^2 \prod_{i\ne m} dt_i}{\int F^2},$$

the supremum over square-integrable $F \not\equiv 0$ supported on $L\mathcal{R}_k$. Then $M_{k,\varepsilon} = M_k(1+\varepsilon, 1-\varepsilon)$ and $M_k = M_k(1,1)$. The quantity $M_k(L,\tau)$ is non-decreasing in L and in τ, satisfies $M_k(cL, c\tau) = c\, M_k(L,\tau)$, and is non-decreasing in k by the embedding used in the proof of Proposition 10. Write $S = t_1 + \cdots + t_k$ and call the index m *active* at t if $S - t_m \le \tau$.

Lemma 13 (Proved here; it is [14, Lemma 6.1] with the region of the marginals kept)**.** *Let $\psi_1, \dots, \psi_k$ be measurable functions on $L\mathcal{R}_k$, positive almost everywhere, such that for almost every $t' \in \tau\mathcal{R}_{k-1}$ and every m*

$$\int_0^{L-|t'|} \frac{du}{\psi_m(t', u)} \le 1,$$

where (t', u) is the point whose m-th coordinate is u and whose other coordinates are t'. Then

$$M_k(L, \tau) \le \operatorname*{ess\,sup}_{t \in L\mathcal{R}_k} \sum_{m \text{ active at } t} \psi_m(t).$$

Proof. For $t' \in \tau\mathcal{R}_{k-1}$ the Cauchy–Schwarz inequality gives

$$\left(\int F \, dt_m \right)^2 \le \int \frac{dt_m}{\psi_m} \cdot \int \psi_m F^2 \, dt_m \le \int \psi_m F^2 \, dt_m.$$

Integrating over $t' \in \tau\mathcal{R}_{k-1}$ gives the integral of $\psi_m F^2$ over the set where m is active. Sum over m. □

Proposition 14 (Proved here; elementary, possibly folklore)**.** *For $k \ge 2$ and $0 \le \varepsilon < 1$,*

$$M_{k,\varepsilon} \le (1+\varepsilon) M_k \le (1+\varepsilon) \frac{k}{k-1} \log k.$$

Conversely, for $0 \le \varepsilon \le M_k/(4k)$,

$$M_{k,\varepsilon} \ge (1+\varepsilon) M_k - \frac{64k^2}{M_k} \varepsilon^2.$$

In particular $\lim_{\varepsilon \to 0^+} (M_{k,\varepsilon} - M_k)/\varepsilon = M_k$.

Proof. By monotonicity in τ and scaling, $M_k(1+\varepsilon, 1-\varepsilon) \le M_k(1+\varepsilon, 1+\varepsilon) = (1+\varepsilon) M_k$, and $M_k \le \frac{k}{k-1} \log k$ is [14, Cor. 6.4].

For the lower bound, scaling gives $M_{k,\varepsilon} = (1+\varepsilon) M_k(1, \rho)$ with $\rho = (1-\varepsilon)/(1+\varepsilon)$. Put $\delta = k(1-\rho)$ and let F on $\mathcal{R}_k$ satisfy $\sum_m J_k^{(m)}(F) \ge (M_k - \eta) I_k(F)$. Split $F = F_1 + F_2$ with $F_2 = F \mathbf{1}_{S > \rho}$, and let $x = I_k(F_2)/I_k(F)$. Two facts follow from the Cauchy–Schwarz inequality on segments of length at most $1-\rho$. First, a fiber of F_2 has length at most $1-\rho$, so $\sum_m J_k^{(m)}(F_2) \le \delta \, I_k(F_2)$. Second, a fiber over t' with $|t'| > \rho$ lies in $\{S > \rho\}$ and has length at most $1-\rho$, so restricting the marginals of F to $\rho\mathcal{R}_{k-1}$ loses at most $\delta \, I_k(F_2)$. By the triangle inequality for $F \mapsto (\int F \, dt_m)_{m \le k}$ and the first fact,

$$M_k - \eta \le \left(\sqrt{M_k(1-x)} + \sqrt{\delta x} \right)^2 \le M_k(1-x) + 2\sqrt{M_k \delta x} + \delta x.$$

This is a quadratic inequality in $\sqrt{x}$. If $\delta \le M_k/2$ it gives $x \le (16\delta + 8\eta)/M_k$. By the second fact the quotient of F for $M_k(1, \rho)$ is at least $M_k - \eta - \delta x$. Letting $\eta \to 0$ we get $M_k(1, \rho) \ge M_k - 16\delta^2/M_k$. Finally $\delta = 2k\varepsilon/(1+\varepsilon)$, and $\delta \le M_k/2$ when $\varepsilon \le M_k/(4k)$. □

So for fixed k the factor $1+\varepsilon$ is exact to first order, and no dual argument can improve it there. Any saving is an effect of large k: it needs ε of size about $1/k$ or more, where the shell $\rho < \sigma \le 1$ carries most of the mass. The first inequality already restricts the parameter: with Bombieri–Vinogradov the criterion $M_{k,\varepsilon} > 4$ forces $\varepsilon > \varepsilon_0(k) = \frac{4(k-1)}{k \log k} - 1$ (Computed):

k	49	48	45	40	39	35	30
$\varepsilon_0(k)$	0.0068	0.0117	0.0274	0.0572	0.0638	0.0929	0.1369

For larger ε we use the lemma with a weight that changes shape where the indices start to become inactive.

Theorem 15 (PROVED HERE for the inequality; COMPUTED, interval arithmetic, for the numerical statement)**.** *Let $k \geq 2$, $0 < \tau < s_1 \leq L$ and $a \geq s_1$. Then*

$$M_k(L,\tau) \leq \sup_{0\leq\sigma\leq\tau} W(\sigma), \qquad W(\sigma) = \frac{ka}{k-1}\log\left(1+\frac{(k-1)(s_1-\sigma)}{a-\sigma}\right) + L - s_1 + \sigma\log\frac{L-\sigma}{s_1-\sigma}.$$

Consequently $M_{k,\varepsilon} < 3.99$ for every $k \leq 29$ and every $0 \leq \varepsilon < 1$. In particular the criterion of [14, Thm. 3.12] *on the ε-enlarged simplex cannot give* DHL$[k,2]$ *for any $k \leq 29$ from the Bombieri–Vinogradov theorem, nor from any level of distribution $\theta \leq 0.501$, whatever the value of ε.*

Proof. Fix $\Lambda > 0$, put $\lambda = \Lambda/(ka)$ and take the two-zone weight

$$\psi_m(t) = \lambda\,(a - S + kt_m) \quad (S \leq s_1), \qquad \psi_m(t) = \Lambda\,\frac{t_m}{S} \quad (S > s_1).$$

It is positive wherever $t_m > 0$. Where $S \leq s_1 \leq a$ every term is non-negative, so the sum over the active indices is at most the sum over all indices, which is $\lambda(k(a-S)+kS) = \Lambda$. Where $S > s_1$ the sum over the active indices is at most $\Lambda\sum_m t_m/S = \Lambda$. On the fiber over t' with $|t'| = \sigma \leq \tau$ we have $S = \sigma + u$, and

$$\int_0^{s_1-\sigma} \frac{du}{\lambda(a-\sigma+(k-1)u)} + \int_{s_1-\sigma}^{L-\sigma} \frac{(\sigma+u)\,du}{\Lambda u} = \frac{W(\sigma)}{\Lambda}.$$

Lemma 13 with $\Lambda = \sup W$ gives the inequality.

For the numerical statement it suffices to treat $k = 29$. For $\varepsilon \leq 143/1000$, Proposition 14 gives $M_{29,\varepsilon} \leq 1.143 \cdot \frac{29}{28}\log 29 < 3.9863$. The interval $[143/1000, 1]$ is cut into 857 intervals $[e, e']$ of length $1/1000$. For each one the file `code/cert_floor_k29.json` lists rational a and s_1 with $\sup W < 3.99$ for $(L,\tau) = (1+e', 1-e)$; by monotonicity in L and τ this covers every $\varepsilon \in [e, e']$. The supremum over σ is bounded rigorously: the first logarithm is non-increasing in σ (because $a \geq s_1$) and the last term is non-decreasing, so on a piece $[x, y]$ we bound W by the first term at x plus the last at y, in ball arithmetic at rational points, bisecting until the bound is below 3.99. The script is `code/dual_eps_cert.py`; a second script, `code/check_floor_cert.py`, re-checks the certificate with a different interval library. The largest value of $\sup W$ over the 857 intervals is 3.9772 (floating point). □

Remark 16 (what was known, and a caution). Polymath8b state the uniform bound $M_{k,\varepsilon} \leq \frac{k}{k-1}\log(2k-1)$ for $0 \leq \varepsilon < 1$ [14, Prop. 6.5], and remark that the bounds for $M_{k,\varepsilon}$ "do not appear to be as sharp". That bound is below 4 exactly for $k \leq 23$, so Theorem 15 moves this floor from 24 to 30. It does not use their bound. We note, with caution, that we could not complete the last step of the proof of [14, Prop. 6.5] as we read it: the weights $1 - S + kt_m$ are summed, and for $S > 1$ their sum over the active indices can exceed k (at $t = (s, 0, \ldots, 0)$ with $1 < s \leq 1+\varepsilon$ only the first index is active and its weight is $1 + (k-1)s$). Our reconstruction gives the constant $\frac{k}{k-1} + \varepsilon$ in place of $\frac{k}{k-1}$. For version 0.4 we re-read the proof in the current arXiv text: after the Cauchy–Schwarz step it says only "Integrating in $t_1, \ldots, t_{i-1}, t_{i+1}, \ldots, t_k$ and summing in i, we obtain the claim." The weights $1 - S + kt_i$ of the inactive indices can be negative when $S > 1$, and dropping them is what lets the sum exceed k; it is at most $k + (k-1)(S-1) \leq k + (k-1)\varepsilon$, which gives our constant. So the argument as printed proves $M_{k,\varepsilon} \leq \left(\frac{k}{k-1} + \varepsilon\right)\log(2k-1)$. Whether the stated inequality is nevertheless true is OPEN; we have not seen the journal version, and we may be missing an intended step. A brief search (the paper [14] and the Polymath wiki page on the variational problem) found no sharper published upper bound for $M_{k,\varepsilon}$; the search was not exhaustive, and Theorem 15 has not been reviewed by anyone outside this project.

Remark 17 (COMPUTED, floating point only: how far this weight goes). The best value of $\sup W$ over (a, s_1), maximised over ε, is about 3.975 for $k = 29$, 4.004 for $k = 30$, 4.257 for $k = 40$ and 4.459 for $k = 50$, in each case near $\varepsilon = 0.24$ to 0.28; the bound of [14, Prop. 6.5] is 4.187, 4.218, 4.482 and 4.689. For small ε the two-zone weight does not beat the scaling bound: at $k = 50$ and $\varepsilon = 1/25$ both give 4.15 (4.1458 against 4.1515), while Polymath8b's certified lower bound is 4.0043. The true threshold is presumably close to $k = 50$, so the theorem is far from sharp. Script: `code/dual_eps.py remark15`.

Remark 18 (a weight that did not work). We also tried the weight $\psi_m = \frac{\beta}{n}(a - S_A + nt_m)$, where n is the number of active indices at t and S_A the sum of the active coordinates. Its sum over the active indices is exactly βa everywhere, so the bound is a times the largest fiber integral. Evaluating the fiber integral exactly on the fibers over $t' = (x, \dots, x, 0, \dots, 0)$ (COMPUTED, floating point, $k = 50$, $\varepsilon = 1/25$, every $a \in [1.05, 3]$ on a grid of step 0.01, script `code/dual_eps.py remark16`) shows that this weight cannot give less than 4.178, which is worse than the 4.1515 from scaling. The binding fiber has a single non-zero coordinate $x \approx 2\varepsilon$: there the zero coordinates become inactive while n is still large, and the normalisation by n wastes the slack. A dual bound that is sharper for small ε must treat the small coordinates near such sparse points differently. This is OPEN. (An earlier draft of this remark, version 0.3 as first circulated, reported a weaker obstruction, 4.146, from fibers with about ten non-zero coordinates; the exact fiber evaluation above supersedes it.)

The limit of weights that see only t_m and S (version 0.4). Every weight used so far has the form $\psi_m(t) = \Psi(S - t_m, t_m)$: Polymath8b's $\lambda(a - S + kt_m)$ and both zones of Theorem 15. Call this family $\mathcal{W}$. For weights in $\mathcal{W}$ the problem "minimise the bound of Lemma 13" is convex: each fiber condition $\int du/\Psi \le 1$ is convex in Ψ, and each point t gives a linear condition $\sum_{m \text{ active}} \Psi(S - t_m, t_m) \le \Lambda$. So its optimum can be computed, and it tells us how far any weight in $\mathcal{W}$, however clever, can go.

Proposition 19 (COMPUTED, floating point, discretised; not a rigorous bound)**.** *Discretise $\Psi = \rho \cdot (L - \sigma + (k-1)u)$ with ρ constant on cells of an $N \times$ (geometric) grid in (S, u), impose the fiber conditions exactly on the grid $\sigma \in \frac{L}{N}\mathbb{Z}$, and impose the pointwise condition on all configurations of the grid of step L/N (found by a knapsack oracle). The convex optimum $B_N(k, \varepsilon)$, computed by cutting planes with an exact convex solver at each step, is:*

(k, ε)	$N = 80$	$N = 120$	$N = 160$	*best weight known*	$M_{k,\varepsilon} \ge$
$(50, 0)$	*3.9919*			*3.9919 (Cor. 6.4)*	*3.9274*
$(50, 1/25)$	*4.1427*	*4.1420*	*4.1418*	*4.1458 (Thm. 15)*	*4.0012*
$(30, 1/4)$	*4.0501*	*4.0345*	*4.0240*	*4.0021 (Thm. 15)*	–

At $\varepsilon = 0$ the optimum over the whole family is the classical weight: the solver, started anywhere, returns $\frac{k}{k-1}\log k$. At $(50, 1/25)$ the values have settled near 4.14: no weight that sees only t_m and S can bring the dual bound much below 4.14 there, while the true value is at least 4.0012. At $(30, 1/4)$ the discretisation has not converged (the values still fall, and lie above the continuum value ≤ 4.0021 that Theorem 15 already certifies), so we draw no conclusion about the exact floor of $\mathcal{W}$; runs at $k = 31, 32$ gave $B_{120} = 4.0261$ to 4.0976, also unconverged. The lesson is the first one: *to approach the truth near $k = 50$, a dual weight must depend on more of t than t_m and S.* The optimal weight is known in principle: if $F > 0$ is the extremal function, $\psi_m = \int F\, dt_m/F$ makes Lemma 13 an equality (the Collatz–Wielandt principle for the positive operator $F \mapsto \sum_{m \text{ active}} \int F\, dt_m$). A first test (COMPUTED, floating point, 20000 random points, not a bound; `code/cw_sample.py`) is encouraging on the plain simplex and discouraging on the enlarged one. For $k = 3$, $\varepsilon = 0$ and the degree-12 Galerkin function ($M_3 \ge 1.646440$) the sampled

ratio stays in $[1.6464, 1.6465]$, which would be an almost sharp upper bound if the supremum could be certified. For $\varepsilon > 0$ the Galerkin polynomial changes sign ($\min F / \max F = -0.0003$ at $(k, \varepsilon, d) = (10, 1/5, 12)$, -0.11 at $(50, 1/25, 12)$). This is structural: where no index is active the extremal function vanishes, so near that region any positive F makes the ratio large. Making the principle usable there is OPEN. The script is `code/dual_lp.py`; `runs/dual_lp_conv.log` and `runs/dual_lp_floor.log` hold the output. The last column is from Table 2 and Theorem 6.

Remark 20 (heuristic: does the enlarged simplex stop at $k = 50$?). The certified values of Table 2 grow with the degree roughly geometrically. Fitting $M(D) = M_\infty - Cr^D$ to three consecutive even degrees (Aitken's Δ^2, script `code/extrapolate_eps.py`) can be tested where the answer is known. From the COMPUTED values at $D = 16, 18, 20$ for $k = 50$, $\varepsilon = 1/25$ it predicts 3.99324, 4.00195, 4.00553 at $D = 22, 25, 27$, against the certified 3.99310 (degree 22), our 4.00125 (degree 25) and Polymath8b's 4.0043 (degree 27). The model is slightly optimistic, by 0.0001 to 0.0012. Applied to $k = 49$ with $D = 18, 20, 22$ (certified 3.97715 and 3.97690 at degree 22 for $\varepsilon = 1/30$ and $1/25$) it gives limits 3.9922 and 3.9938. As a test registered before the run, the fit predicted 3.98230 at degree 24 for $\varepsilon = 1/30$; the certified value is 3.98218 (1508 coefficients), again slightly below, and the fit through $D = 20, 22, 24$ moves the limit down to 3.9913. The grid $\varepsilon \in \{1/50, 1/40, 1/30, 1/25, 1/20\}$ at $D \le 20$ gives at most 3.995. This suggests that $\sup_\varepsilon M_{49,\varepsilon} < 4$, so that the ε-enlarged simplex with Bombieri–Vinogradov stops at $k = 50$, consistent with the fact that every bound below 246 uses a different support. This is a heuristic, not a theorem: OPEN. Together with Theorem 15 it brackets the floor of this method between 30 and (conjecturally) 50.

Remark 21 (scope). Theorem 15 concerns the ε-enlarged simplex only. It says nothing about the 2026 claims, which use other supports and new equidistribution estimates. A floor for those is the Problem in the next section.

5. Stadlmann's support (versions 0.5 and 0.6)

This section takes the first steps on the next rung, Stadlmann's region for $H_1 \le 240$ [18], and gives a first rigorous answer to the Problem in the Programme (Section 6) for the one-band supports used by [18] and [3]. For this version a helper re-read both sources (the arXiv text of [18], the draft [3]); our notes with page references are in `notes/stadlmann-recheck-s5.md`.

The support and the criterion. The following is LITERATURE, read in [18, Def. 1, Def. 5, Prop. 1]. For $\delta > 0$, $-\varepsilon = A_0 < A_1 < \cdots < A_n < \frac{1}{2} - \varepsilon$ and caps $\delta < B_{j,m} \le B_{j,m+1} \le B_{j,m} + \delta$, the support $T_k(\delta, \underline{A}, \underline{B}, \varepsilon)$ is the set of $t \in [0,1]^k$ such that, for the j with $\sum_i t_i \in [A_{j-1}+\varepsilon, A_j+\varepsilon)$, one has $\sum_{i \in I} t_i \le B_{j,|I|}$ where $I = \{i : t_i > \delta\}$ (the "large" coordinates). With one band ($n = 1$), the functional $J(F)$ of [18] is the integral of $\left(\int F\, dt_k\right)^2$ over $t_1 + \cdots + t_{k-1} \le A_1 - \varepsilon$ (our reduction of the general definition), $I(F) = \int_{T_k} F^2$, and the criterion is $k(1-c_1)J(F) - kc_2K(F) > I(F)$ for a symmetric F supported on T_k, where $c_1, c_2 \ge 0$ come from the minorant and $K(F) \ge 0$. Stadlmann's choice is $k = 49$, $\varepsilon = 0.0075$, $A_1 = 0.253$, $\delta = 0.028$, $B_{1,1} = B_{1,2} = 0.15$, $B_{1,m} = 0.17$ ($m \ge 3$) and $c_1 = c_2 = 0$; that of [3] is $k = 45$, $\varepsilon = 1/125$, $A_1 = 257/1000$, $\delta = 41/2500$ with caps listed in their Table 3. Both use one band. The equidistribution estimates behind the criterion are CLAIMED; we have not checked them.

In the Polymath scaling $u = 4t$ the support lies in $L\mathcal{R}_k$ with $L = 4(A_1 + \varepsilon)$, the marginals are integrated over $\tau\mathcal{R}_{k-1}$ with $\tau = 4(A_1 - \varepsilon)$, and the criterion with $c_1 = c_2 = 0$ reads $kJ/I > 4$. Write $s = (L+\tau)/2 = 4A_1$ and $e = (L-\tau)/(L+\tau) = \varepsilon/A_1$; the scaling $M_k(cL, c\tau) = cM_k(L, \tau)$ of Section 4 gives

$$M_k(L, \tau) = s\, M_k(1+e, 1-e) = 4A_1\, M_{k,\varepsilon/A_1}. \tag{2}$$

For Stadlmann, $(L,\tau) = (1.042, 0.982)$, $s = 253/250$, $e = 15/506$; for [3], $(L,\tau) = (1.06, 0.996)$, $s = 257/250$, $e = 8/257$.

Without the caps.

Proposition 22 (Computed, exact rational arithmetic)**.** *With even power sums, $M_{49,15/506} \geq 3.9685535$ (degree 20, 707 coefficients) and $M_{45,8/257} \geq 3.9029887$ (degree 21, 846 coefficients). Hence, by (2), $M_{49}(1.042, 0.982) \geq 4.016176$ and $M_{45}(1.06, 0.996) \geq 4.012272$.*

The two certificates, in the directory `code/`, are `cert_stad49_capfree_D20e.json` and `cert_ax45_capfree_D21e.json`; the script `verify_cert.py` re-checks them. Products of even power sums and even-part monomial symmetric polynomials span the same space, so our degree-21 basis is the space $\mathcal{B}_{21}$ of [3, Thm. 11.1]; its dimension, 846, agrees with theirs. On their capped support they report the exact value $4.0043840983\ldots$ (Claimed); on the larger cap-free region, with the same space, we get 4.012272. Both are lower bounds, for M_T and for $M_{45}(1.06, 0.996) \geq M_T$ respectively, so the comparison proves nothing; but it suggests (if their value is right) that at this degree the caps cost about 0.008. This is a comparison, not a verification of [3].

With the caps. For $F = P\,1_T$ with P a polynomial, write $I(P)$, $J(P)$ for the cap-free functionals on $L\mathcal{R}_k$ (marginals over $\tau\mathcal{R}_{k-1}$) and $V = L\mathcal{R}_k \setminus T$ (Polymath scaling; $g = 4\delta$, $C_m = 4B_{1,m}$).

Proposition 23 (Proved here; elementary)**.** *Let $R_m(g,c) = \{u \in L\mathcal{R}_k : \ u_1, \ldots, u_m > g,\ u_1 + \cdots + u_m \geq c\}$, put $c_m = C_m$ for $m \leq 6$ and $c_7 = 7g$. Then, with*

$$E(P) = \sum_{m=1}^{7} \binom{k}{m} \int_{R_m(g,c_m)} P^2,$$

one has $I_T(F) \leq I(P)$ and, for every $0 < \eta < 1$,

$$J_T(F) \ \geq \ \max\Big(J(P) - 2\sqrt{J(P)\,L\,E(P)},\ (1-\eta)J(P) - (\eta^{-1}-1)L\,E(P) \Big).$$

Proof. A point of V has a large set I with $\sum_{i\in I} u_i > C_{|I|}$. If $|I| \leq 6$ it lies in a copy of $R_{|I|}(g, c_{|I|})$; if $|I| \geq 7$, any seven of its large coordinates exceed g and sum to more than $7g$, so it lies in a copy of $R_7(g, c_7)$. By symmetry of P^2, $\int_V P^2 \leq E(P)$. On the fiber over u' write $a = \int P\,du_k$ and $b = \int P\,1_V\,du_k$, so that the marginal of F is $a - b$. Cauchy–Schwarz gives $b^2 \leq L \int P^2 1_V\,du_k$, hence $\int b^2 \leq L\,E(P)$. Then $(a-b)^2 \geq a^2 - 2|a||b|$ with Cauchy–Schwarz once more, or $(a-b)^2 \geq (1-\eta)a^2 - (\eta^{-1}-1)b^2$. Finally $I_T(F) = \int_T P^2 \leq I(P)$. □

The integrals over $R_m(g,c)$ are again exact rational numbers for the basis $(L - P_1)^a P^b$ (Proved here; `code/caps_exact.py`). Write the first m coordinates as $g + z$. Integrating out the other $k - m$ coordinates with Lemma 3 leaves powers of $L - mg - |z|$; the shifted power sums are $P_j(g+z) = \sum_l \binom{j}{l} g^{j-l} P_l(z)$ with $P_0 = m$ and $P_1 = |z|$; at fixed $r = |z|$ the integral of $\prod_l P_l(z)^{f_l}$ is $r^{m-1+w(f)} E_m(f)/(m-1+w(f))!$ (now with odd parts as well); and the remaining integral in r over $[\max(c - mg, 0), L - mg]$ is elementary. The code reproduces Lemma 3 exactly when $g = c = 0$, and agrees with direct symbolic integration (sympy) for $k = 3$, $m = 1, 2$, $g = 1/10$.

Since all three quantities are exact quadratic forms in the coefficients of P, we choose P as the top eigenvector of $(1-\eta)J - (\eta^{-1}-1)L\,G_E$ against I, where G_E is the Gram matrix of E, and evaluate both bounds exactly. The penalty works: at degree 16 it lowers E/I from $3.2 \cdot 10^{-4}$ (for the cap-free optimiser) to $3.4 \cdot 10^{-7}$.

Proposition 24 (COMPUTED, exact rational arithmetic)**.** *For Stadlmann's parameters ($k = 49$) there is a polynomial P of degree 20 in even power sums (707 coefficients) with*

$$\frac{kJ_T(P\,1_T)}{I_T(P\,1_T)} \;\geq\; 3.95972.$$

At degree 16 (307 coefficients) the same method gives 3.93625.

This is below 4, and in version 0.5 the rung 240 was therefore not reproduced; Theorem 26 below supersedes it. The certificate is `code/cert_stad49_caps_D20e.json`, written by `code/stadlmann_caps_opt.py`; the best bound came from $\eta = 1/800$, with $E/I = 1.6 \cdot 10^{-7}$ and the bound of Proposition 23 in its square-root form. Where the loss comes from (COMPUTED, same runs): the cap-free optimiser of degree 20 has $s\,kJ/I = 4.01618$ but $E/I = 1.2 \cdot 10^{-3}$ (of which $1.2 \cdot 10^{-4}$ from points with one coordinate above 0.6); forcing E down costs about 0.045 of the cap-free ratio ($4.0161 \to 3.9710$), and the Cauchy–Schwarz step costs another 0.011. Both losses come from the way we bound J_T, not from the caps as such: $F = P\,1_T$ is cut off on V anyway, and Proposition 23 charges for P on V only because it does not compute the capped marginals. The next step is therefore an exact formula for $J_T(P\,1_T)$, which needs the fibers of T and not only integrals over V. A first attempt that dropped the condition $u_i > g$ in the union bound failed badly ($E/I \approx 146$ for the cap-free optimiser; `runs/stad49_caps_D20.log`).

The capped functionals exactly (version 0.6). Version 0.5 bounded $J_T(P\,1_T)$ through Proposition 23, and the Cauchy–Schwarz step there turned out to be the whole difficulty. We now compute $I_T(P\,1_T)$ and $J_T(P\,1_T)$ exactly. Notation as above: u in the Polymath scaling, $R = L - |u'|$ on the fiber over $u' \in \mathbb{R}^{k-1}_{\geq 0}$, $I(u) = \{i : u_i > g\}$, $m(u) = |I(u)|$, $s(u) = \sum_{i \in I(u)} u_i$, and $Q(u', x) = \int_0^x P(u', t)\,dt$.

Proposition 25 (PROVED HERE; elementary)**.** *Assume $C_{m+1} \leq C_m + g$ for all m (this is Stadlmann's condition $B_{j,m+1} \leq B_{j,m} + \delta$). Call u'* admissible *if $s(u') \leq C_{m(u')}$. Then:*

1. *the fiber $\{t \geq 0 : (u', t) \in T\}$ is empty if u' is not admissible, and otherwise equals $[0, U(u')]$ with $U = \min\big(R, \max(g,\, C_{m(u')+1} - s(u'))\big)$;*
2. *one has $I_T(P\,1_T) = I(P) - \int_V P^2$ and*
$$J_T(P\,1_T) = J(P) - \mathrm{Corr}_1 - \mathrm{Corr}_2,$$
where $\mathrm{Corr}_1 = \int Q(u', R)^2$ over the non-admissible u' with $|u'| \leq \tau$, and $\mathrm{Corr}_2 = \int \big(Q(u', R)^2 - Q(u', U)^2\big)$ over the admissible u' with $U < R$;
3. *each of the three correction integrals is a finite signed sum, over the size m of the large set and over $j \geq 0$ further coordinates forced above g (inclusion–exclusion for "all other coordinates are at most g", with weight $(-1)^j \binom{n}{m}\binom{n-m}{j}$), of integrals over polygons in the plane of $r_A = s - mg$ and $\rho = |u| - s - jg$. On each polygon the integrand, after the shape integrals of Lemma 3 and of Proposition 23, is a polynomial in (r_A, ρ), so every term is an exact rational number.*

For Stadlmann's parameters every correction piece has $m \leq 9$ and $j \leq 3$.

Proof. (1) If $t \leq g$ the large set of (u', t) is $I(u')$, so $(u', t) \in T$ iff u' is admissible and $|u'| + t \leq L$. If $t > g$ the large set is $I(u') \cup \{k\}$ and the condition is $s(u') + t \leq C_{m(u')+1}$; since $C_{m+1} \leq C_m + g$, this condition with $t > g$ implies $s(u') < C_{m(u')}$, so a non-admissible u' has an empty fiber. Hence for admissible u' the fiber is $[0, \min(R, g)] \cup (g, \min(R, C_{m+1} - s)]$, which is $[0, U]$. (2) $F = P\,1_T$ has marginal $Q(u', U)$ on admissible u' and 0 otherwise, while the cap-free marginal is $Q(u', R)$; subtract. (3) By symmetry, integrate over the points whose large set is exactly $A = \{1, \ldots, m\}$ and multiply by $\binom{n}{m}$; the condition that the remaining coordinates are at most g is $\prod(1 - 1_{u_i > g})$,

expanded. On such a piece put $z = u - g$ on the $m + j$ forced coordinates; at fixed $r_A = |z_A|$ and $\rho = |z_B| + |y|$ (y the free coordinates) the integral of a product of power sums is a sum over splits of the product of the A-group density of Proposition 23 and the convolution of the j shifted coordinates with the free ones, which is again of Dirichlet type. The integrand (P^2, $Q(R)^2$, $Q(g)^2$ or $Q(x)^2$ with $x = C_{m+1} - s$) is a polynomial in R and x, both affine in (r_A, ρ). The regions are: $r_A \geq C_m - mg$ and $r_A + \rho \leq$ const (the non-admissible pieces); $r_A \geq C_{m+1} - (m+1)g$ and $R > g$ (where $U = g$); and the rectangle $r_A < C_{m+1} - (m+1)g$, $\rho < L - C_{m+1} - jg$ (where $U = C_{m+1} - s$). In every piece the large set carries at least $C_1 = 0.6$ or the free mass is below $L - C_1$, so $jg < 0.442$, $j \leq 3$; and $mg < L$ gives $m \leq 9$. Both counts are LEAN-CHECKED (`stadlmann_extra_le_three`, `stadlmann_large_count`). □

Theorem 26 (COMPUTED, exact rational arithmetic, two independent programs)**.** *Let P be the cap-free optimiser of degree* 20 *of Proposition 22 (*707 *coefficients, even power sums). For Stadlmann's parameters ($k = 49$, $L = 1.042$, $\tau = 0.982$, $g = 0.112$, $C_1 = C_2 = 0.6$, $C_m = 0.68$ for $m \geq 3$) and $F = P\,1_T$,*

$$\frac{k\,J_T(F)}{I_T(F)} = 4.0120389035\ldots \;>\; 4.$$

The polynomial, in u-coordinates, is `code/cap6/P_capfree_D20_u.json` (written from the certificate `code/cert_stad49_capfree_D20e.json` by `code/to_ucoords.py`; the cap-free value kJ/I it gives equals $\frac{253}{250}$ times the certified Rayleigh quotient exactly). Two programs compute I_T and J_T as exact rationals and agree on both numbers digit for digit: `code/capped_exact.py` (ours; Proposition 25, "cap-free minus corrections", 46 minutes on three cores) and `code/capped_check.py`, written by a helper from the definitions only, without access to our code, which sums directly over the exact large set (no cap-free term at all) with exponential-moment generating functions (6 minutes on four cores). The checks behind them: the direct and the corrected I_T in `capped_exact.py` agree exactly; a Monte Carlo test at $k = 12$, $D = 4$, where the caps remove 47% of I and 45% of J, agrees to within one standard deviation (`code/cap6/mc_test.py`); and `capped_check.py` agrees exactly with a brute-force evaluation by polytope triangulation and symbolic integration for $k = 2, \ldots, 5$ with caps that bite (`code/capped_check_tests.py`). Results: `code/cap6/res_capfree_D20.json`, `runs/capped_check_D20.json`.

The caps cost only 0.0041: relative to the cap-free values, $\int_V P^2 = 1.209 \cdot 10^{-3}\, I(P)$, and Corr_1, Corr_2 on the pieces $U = g$ and $U = C_{m+1} - s$ are $1.72 \cdot 10^{-4}$, $8.1 \cdot 10^{-5}$ and $1.98 \cdot 10^{-3}$ of $J(P)$. The largest loss is the cut of the fiber at $C_{m+1} - s$ (for $m = 0$: no coordinate above 0.6). So Proposition 24 (3.9597) measured the weakness of Proposition 23, not the caps; it is superseded. The loss grows with the degree of the cap-free optimiser (COMPUTED, exact, `capped_check.py`; degrees 8, 12 and 16 also with `capped_exact.py` at degree 8):

degree	8	12	16	20
cap-free kJ/I	3.789069	3.923943	3.987180	4.016176
capped kJ_T/I_T	3.789068	3.923849	3.985976	4.012039

so higher-degree optimisers put more mass near the caps; optimising P on T itself (not done yet) can only raise the capped value. At $k = 45$ this is consistent with the comparison after Proposition 22, where the caps of [3] appeared to cost about 0.008.

Corollary 27 (conditional; the analytic input is CLAIMED)**.** *The variational step of* [18] *is independently reproduced: the criterion of* [18, Prop. 1] *holds for $k = 49$ with his support, his caps and his parameters, and $c_1 = c_2 = 0$. Together with the* LEAN-CHECKED *admissible* 49*-tuple*

of diameter 240 *(Theorem 1),* $H_1 \le 240$ *then follows from his Prop. 1 and his equidistribution theorem, which we have not verified.*

This is a reproduction of part of a claimed result, not a new bound, and it says nothing about the correctness of the new equidistribution estimates of [18] (his Lemmas 3–7 and Prop. 3), on which everything rests. Our formulation agrees with [18] as re-read in our notes (`notes/stadlmann-recheck-s5.md`), with these differences, none of which changes the value: we use the scaling $u = 4t$ (criterion $kJ/I > 4$); we use our own reduction of his J for one band (the band conditions on t_k and t_k' follow from the support of F); we ignore his closed versus half-open band endpoints (measure zero); and we use our own test space (even power sums times powers of $L - P_1$, degree 20) instead of his $\mathcal{B}_{19}$ or $\mathcal{B}_{21}$, whose coefficients he does not print.

Remark 28 (Computed, interval arithmetic: a ceiling, our own). Every coordinate of a point of T is at most 0.6 (Lean-checked: `stadlmann_coord_le`) and $T \subset L\mathcal{R}_k$, so the capped quantity is at most $M_{49}(1.042, 0.982)$, which Theorem 15 with $a = 1.0417$, $s_1 = 1.0007$ bounds by 4.14 (certified with `dual_eps_cert.certify_box`, 337 pieces). Hence $4.0120 < \sup_F kJ_T/I_T \le 4.14$. Cutting the fibers of the two-zone weight at 0.6 lowers its floating-point ceiling only from 4.13785 to 4.13753: the supremum of W sits near $\sigma = \tau$, where the fibers are short anyway, so a ceiling that sees the caps needs a weight that sees the large coordinates (Open).

A floor for one-band hybrid supports (our own). Polymath8b's ε-simplex is the case $A_1 = 1/4$ without caps, so the following extends Theorem 15 to every one-band support of Stadlmann's type.

Theorem 29 (Proved here for the reduction; Computed, interval arithmetic, for the table)**.** *Let* $T = T_k(\delta, (-\varepsilon, A_1), \underline{B}, \varepsilon)$ *be a one-band support with* $\varepsilon \ge 0$ *and* $A_1 \le \frac{1}{4} + \omega$*, with any* δ *and any caps. If the criterion* $k(1 - c_1)J(F) - kc_2K(F) > I(F)$ *holds for some* F *supported on* T*, then* $M_k\left(4(\frac{1}{4} + \omega + \varepsilon), 4(\frac{1}{4} + \omega - \varepsilon)\right) > 4$*. This is false, for every* $0 \le \varepsilon < \frac{1}{4} + \omega$*, in each of the following cases:*

ω	0	3/1000	1/300	7/1000	1/68
no k *with*	$k \le 29$	$k \le 28$	$k \le 28$	$k \le 26$	$k \le 22$

So, whatever δ*, the caps and the minorant, a one-band support cannot give* DHL$[k, 2]$ *for* $k \le 28$ *at Stadlmann's level (*$\omega = 0.003$*; the helper's reconstruction of his conditions gives* $\omega \le 0.0032$*), for* $k \le 26$ *at the level* $\omega = 0.007$ *of* [3]*, and for* $k \le 22$ *whenever* $\omega \le 1/68$*, a range containing every* ω *that satisfies the Type I condition* $68\omega + 14\delta < 1$ *of* [18, Lemma 5].

Proof. As $c_1, c_2 \ge 0$ and $K(F) \ge 0$, the criterion implies $kJ(F) > I(F)$. In the scaling $u = 4t$, F becomes a function supported on $L\mathcal{R}_k$ (because $T \subseteq \{\sum t_i \le A_1 + \varepsilon\}$), the ratio is multiplied by 4, and J is the marginal functional over $\tau\mathcal{R}_{k-1}$; so $4 < M_k(4(A_1 + \varepsilon), 4(A_1 - \varepsilon))$, and monotonicity in L and τ replaces A_1 by $\frac{1}{4} + \omega$. (For $\varepsilon < 0$ one has $\tau > L$ and the quantity is at most LM_k, which is covered by the case $\varepsilon = 0$ below.) Monotonicity in k reduces each entry to the largest k shown. For the numerical part, put $A = \frac{1}{4} + \omega$. When $4(A + \varepsilon)\frac{k}{k-1}\log k < 4$ we use $M_k(L, \tau) \le M_k(L, L) = LM_k \le L\frac{k}{k-1}\log k$ [14, Cor. 6.4]. The remaining range of ε, up to A, is cut into intervals of length at most $1/2000$; on $[e, e']$ we take $(L, \tau) = (4(A + e'), 4(A - e))$, which dominates by monotonicity, and certify $\sup_\sigma W(\sigma) < 4$ with rational (a, s_1) exactly as in the proof of Theorem 15. The certificates are `code/cert_floor_hybrid_k*_w*.json`, written by `code/floor_hybrid.py` in ball arithmetic (`python-flint`) and re-checked by the same script with `--check` in a different interval library (`mpmath.iv`). □

Remark 30 (how sharp). For each ω in the table the same weight fails at the next k: the floating-point $\sup W$ reaches 4 on some interval of ε (for example $k = 27$, $\omega = 0.007$, near $\varepsilon \approx 0.0548$).

So the table shows the reach of this weight, not of the method; the true floor is presumably much higher, as Proposition 22 suggests (it shows that the cap-free quantity already exceeds 4 at $k = 49$ and $k = 45$ for the two parameter sets). The theorem ignores the caps entirely, which only shrink the support; a floor that sees the caps is the natural next step. It also does not apply to supports with several bands, where J contains cross terms between bands, nor to the constructions behind 186 and 182, which we have not read in enough detail.

6. Programme

Track A: lowering the bound.

(1) *Climb the ladder.* The certified pipeline now stands at $k = 50$ ($H_1 \le 246$, the established bound, via the ε-enlarged simplex). Stadlmann's region ($k = 49$, 240): version 0.6 reproduces the variational step exactly (Theorem 26, Corollary 27), so the project's bound is 240 *conditionally on the unverified equidistribution estimates of* [18]. Next rungs: 236 ($k = 48$, [8, 17]) and 212 ($k = 45$, [3], whose caps vary with m; Proposition 25 applies with the obvious changes), then 186 and 182.

(2) *Floor of the method (our main original direction).* For the plain simplex the dual bound $M_k \le \frac{k}{k-1} \log k$ shows where the method stops ($k \ge 51$). For the ε-enlarged simplex Theorem 15 gives $k \ge 30$, which is still far from the expected truth near 50; Section 4 explains what a sharper weight must do, and Proposition 19 indicates that it must depend on more of t than t_m and S (a Collatz–Wielandt weight $\psi_m = \int F\, dt_m / F$ built from a good positive test function F is the natural candidate). For the one-band hybrid supports of [18, 3], Theorem 29 gives the first such bound we know of ($k \ge 29$ at Stadlmann's level, $k \ge 27$ at that of [3]), still far from 49 and 45; it ignores the caps. A rigorous dual certificate there would say whether 182 is close to the limit of the present inputs or far from it, and which input (which Type I/II/III range) is binding.

(3) *Which rung is cheap.* From Table 1, the gain per unit of k below the claimed frontier is $182 \to 176 \to 168 \to 162 \to 158$ for $k = 39, \dots, 35$: the steps $39 \to 38$ and $38 \to 37$ are worth the most.

Problem 31 (Open; partly answered by Theorem 29 for one band, without using the caps)**.** Give a rigorous upper bound for the variational quantity of [18] as a function of k and of the admissible exponents, and deduce the least k the hybrid method can reach with the currently proved estimates.

Track B: the parity barrier. No sieve of the above kind can distinguish integers with an even number of prime factors from those with an odd number. Known ways around the barrier all inject bilinear or structural information. We will study, one per session, what each would need in order to say something about twin primes: Chen's theorem [4]; Bombieri's asymptotic sieve [2]; the parity-breaking sieve of Friedlander and Iwaniec [6]; Heath-Brown's theorem that Siegel zeros would imply the conjecture [7]; the function-field proof of Sawin and Shusterman [16]; and the entropy-based results on correlations of multiplicative functions [19].

Problem 32 (Open, to be made precise in Lean)**.** State, as a single explicit hypothesis on bilinear sums over $n(n+2)$, the weakest known input that implies the twin prime conjecture, and formalise the implication.

Rules of this paper. A numerical bound enters the "established by this project" column only with an exact certificate and, for the combinatorial part, a Lean proof. Ideas are logged with their outcome, including failures. Nothing here is peer reviewed.

References

[1] E. Chen, S. Hariharan, K. Lau, B. Mehta, A. A. Swaminathan, J. Thorner, E. Xie, *Lean formalization of bounded gaps between primes* (blueprint, PrimeGapsLib), 2026. https://primegaps.axiommath.ai/

[2] E. Bombieri, *The asymptotic sieve*, 1976.

[3] F. Charton, L. Hong, K. Lau, K. Ono, G. Remy, H. C. Siu, A. A. Swaminathan, J. Thorner, Y. Xie, *A new bound for small gaps between primes*, preliminary draft, 3 September 2026. https://primegaps.axiommath.ai/bgp212.pdf

[4] J. R. Chen, *On the representation of a larger even integer as the sum of a prime and the product of at most two primes*, 1973.

[5] V. Chvátal, *On certain polytopes associated with graphs*, J. Combin. Theory Ser. B, 1975.

[6] J. Friedlander, H. Iwaniec, *The polynomial $X^2 + Y^4$ captures its primes*, Ann. of Math., 1998.

[7] D. R. Heath-Brown, *Prime twins and Siegel zeros*, Proc. London Math. Soc., 1983.

[8] S. Kintali, *Improved bounded gaps between primes*, preprint, 1 September 2026.

[9] L. Lovász, *Normal hypergraphs and the perfect graph conjecture*, Discrete Math., 1972.

[10] J. Maynard, *Small gaps between primes*, Ann. of Math. 181 (2015), 383–413.

[11] E. W. Weisstein, *Bounded gaps between primes*, MathWorld. https://mathworld.wolfram.com/BoundedGapsBetweenPrimes.html

[12] Nervous Machine, *DEBRIEF-math-primegaps*, a re-run of the repository `openai/PrimeGaps186` by a third party ("Scorecard, not a verification"), status dated 7 November 2026. https://github.com/Nervous-Machine/DEBRIEF-math-primegaps

[13] OpenAI, *Improved short gaps between primes*, preprint dated 30 August 2026, with the repository `openai/PrimeGaps186`.

[14] D. H. J. Polymath, *Variants of the Selberg sieve, and bounded intervals containing many primes*, Res. Math. Sci. 1 (2014).

[15] GPT-6 Astra (OpenAI), prompted by R. Sharifi, *A fixed-profile proof for prime gaps of at most 182*, dated 18 September 2026 (first version 4 September 2026). https://www.math.ucla.edu/~sharifi/AI/prime_gaps_182_expository_bundle.pdf

[16] W. Sawin, M. Shusterman, *On the Chowla and twin primes conjectures over $\mathbb{F}_q[T]$*, Ann. of Math., 2022.

[17] Z. Song, S. Yue, *Bounded gaps between primes: an upper bound of 236*, IACR ePrint 2026/1893.

[18] J. Stadlmann, *Bounded gaps between primes*, arXiv:2608.31126, 31 August 2026.

[19] T. Tao, *The logarithmically averaged Chowla and Elliott conjectures for two-point correlations*, Forum Math. Pi, 2016.

[20] Y. Zhang, *Bounded gaps between primes*, Ann. of Math. 179 (2014), 1121–1174.